

① Smart Home Automation System - Intelligent Agents

A smart home automation system automatically controls lighting, temperature & security based on user behaviour & environmental conditions. It receives inputs from sensors & security cameras.

(i) Simple Reflex Agents
A simple based on the Agent makes decisions only using predefined condition - action rules

Eg: If motion is detected, the lights are turned on

Advantages

- ⇒ Simple & fast
- ⇒ Easy to implement
- ⇒ Suitable for basic automation tasks

Disadvantages

- Cannot remember previous states
- Cannot adapt to user preferences

(ii) Model-Based Agent

⇒ A model-Based agent maintains an internal model of the environment

Eg:

The system remembers which rooms are occupied.

Advantages

- Works well in ~~partially~~ partially observable environment
- Makes better decisions using stored information

Disadvantages

- Requires more memory
- More complex to design & maintain

(iii) Goal - Based Agent
⇒ A Goal - Based Agent selects
that ~~help~~ action that help to
achieve a specific goal.
⇒ It evaluates different possible actions
before making decisions

Eg:
(*) Maintains the room temperature
at 24°C
(*) Ensures all doors are locked
before Bed time

Advantages

- ⇒ Can plan future actions
- ⇒ Flexible & intelligent
- ⇒ Achieves desired goal efficiently.
- ~~##~~

~~Dis~~ Disadvantages

- ⇒ Requires additional computation
- ⇒ Decision-making takes more

time

Conclusion

A hybrid intelligent agent provides the best performance because it can remember past information, plan future actions, choose beneficial decisions. This ensures maximum comfort, energy savings & more security.

②

Problem Statement

A robot in an unknown maze doesn't know ~~the~~ blueprint of the maze. It doesn't know how to exit the maze.

For it, the robot uses sensors to find the obstacle and uses

Uniformed cost search such as UCS
Uniform cost search and IDS
(Iterative deepening search)

Uniform cost Search (UCS)

UCS expands the node having the cost first.
It uses the cost of a particular position as

$$f(n) = g(n)$$

where $g(n)$ is the cost from starting node till current node.

Eg: If two paths are available UCS chooses the path with lowest cost.

Advantages

- ① Complete when step costs are pointers
- ② Finds the least cost solutions

Disadvantages

- ① Requires more memory
- ② Can be slow
- ③ May explore many unnecessary paths

Iterative Deepening

⇒ IDS repeatedly performs depth-limit search with increasing depth limit.

Advantages

- Uses less Memory
- Completes for finite branching
- Finds shortest solution when every step has same cost.

Disadvantages

⇒ Repeat the same nodes during iterations

⇒ Can take additional time

Time & space complexity:

$$\text{Time} = O(b^d)$$

$$\text{Space} = O(bd)$$

Conclusion:

The robot can be considered an intelligent agent because it senses the environment. For a normal maze, a model-based and goal-based agent with IDS is effective.